

Surapaneni Rohan Krishna

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SUMMARY

Python and AI/ML engineer with hands-on experience building **LLM-powered agentic systems**, **deep learning models**, and **RESTful APIs** on AWS. Built a production **ReAct-loop AI agent framework** with multi-provider LLM support, **Retrieval-Augmented Generation (RAG)**, and **vector database** memory from scratch. Published ML research on ensemble **Vision Transformers** for medical image classification using **PyTorch** and **Hugging Face Transformers**. AWS Certified Cloud Practitioner. Seeking a Junior AI/ML Engineer role.

WORK EXPERIENCE

Tecdatum

Mar 2026 – Present

Software Development Intern

- Developed production features using **ASP.NET Core MVC** backend and **React/Bootstrap** frontend; shipped code in an agile team environment.
- Designed and integrated **RESTful APIs**; validated endpoints via **Postman**, reducing integration bugs across the team.

Bharat Dynamics Limited (BDL)

Jun 2024 – Jul 2024

Software Intern

- Built an Online Inspection Memo System using **Java** (DAO pattern) and **MySQL** with role-based access control; integrated backend APIs via a JavaScript/HTML/CSS frontend.
- Gathered requirements from stakeholders and improved system reliability by debugging production issues.

PROJECTS

AgentMemory - ReAct LLM Agent Framework

Python, FastAPI, ChromaDB, OpenAI, Anthropic, Groq

- Built a **ReAct-loop AI agent framework** with multi-provider LLM support (OpenAI, Anthropic, Groq, DeepSeek, OpenRouter), automatic tool-call parsing, and error recovery; exposed via a **FastAPI** REST API with runtime provider switching.
- Integrated **ChromaDB** vector database for **semantic search** and **Retrieval-Augmented Generation (RAG)** over conversation history using **text embeddings** for long-term context retrieval across sessions.

Skin Lesion Classification - Deep Learning Research

PyTorch, Hugging Face Transformers, Python

- Built an ensemble **deep learning** model on the HAM10000 dataset combining **Swin-B Transformer**, **ConvNeXt-B**, and **EfficientNetV2-L** using **PyTorch** and **Hugging Face Transformers**; achieved multi-class **medical image classification** across 7 skin lesion categories.
- Applied conformal prediction for uncertainty quantification; submitted as a **Machine Learning** research publication.

Serverless Blog Platform

React, AWS Amplify, S3, CloudFront, Cognito

- Deployed a fully serverless blog using **AWS Amplify**, **S3**, and **CloudFront** with **Cognito**-based authentication; achieved global CDN delivery with zero server overhead.

SKILLS

Languages: Python, JavaScript, Java, C++

AI / ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Vision Transformers, LLM APIs, ReAct Agent Patterns, Retrieval-Augmented Generation (RAG), Prompt Engineering, Text Embeddings, Semantic Search, ChromaDB, PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy

Frameworks: FastAPI, React, ASP.NET Core MVC, Node.js

Cloud / DevOps: AWS (Lambda, S3, RDS, Cognito, CloudFront, Amplify), Docker, CI/CD, GitHub Actions, Linux, Git

Databases: MySQL, MongoDB, PostgreSQL, ChromaDB

Tools: Postman, Jupyter Notebook, Git, GitHub

EDUCATION & CERTIFICATIONS

B.Tech in Computer Science, SRM University AP

2022 – 2026

CGPA: 6.67/10.0

Class XII, Telangana State Board of Intermediate Education

2022

Score: 715/1000

AWS Certified Cloud Practitioner, Amazon Web Services

2024